IYEC 2026 · SENIOR · TEAM ROOTEDSG

RootStake

A plant that can ask for what it needs, and get it.

Root-zone sensing, solar-powered watering and feeding, and an app that explains it all in one line.



THE MAIN PROBLEM

Home-grown plants die silently. Nobody can see what is happening under the soil until it is too late.

51.2%
of residents stopped growing or never started

36.1%
say they forget or have no time

16.3%
do not know where to look for help



THE OPPORTUNITY

Singapore wants to grow more of its own food

>90%

of food is imported

Singapore depends on supply chains it does not control.

8% → 20%

local vegetables by 2035

Singapore Food Story 2 target, up from about 8% in 2024.

1M+

homes with growing space

Corridors, balconies and windowsills that could each hold a few pots of greens.



THE GAP · 3,043 RESIDENTS SURVEYED

Half of residents want plants they do not have



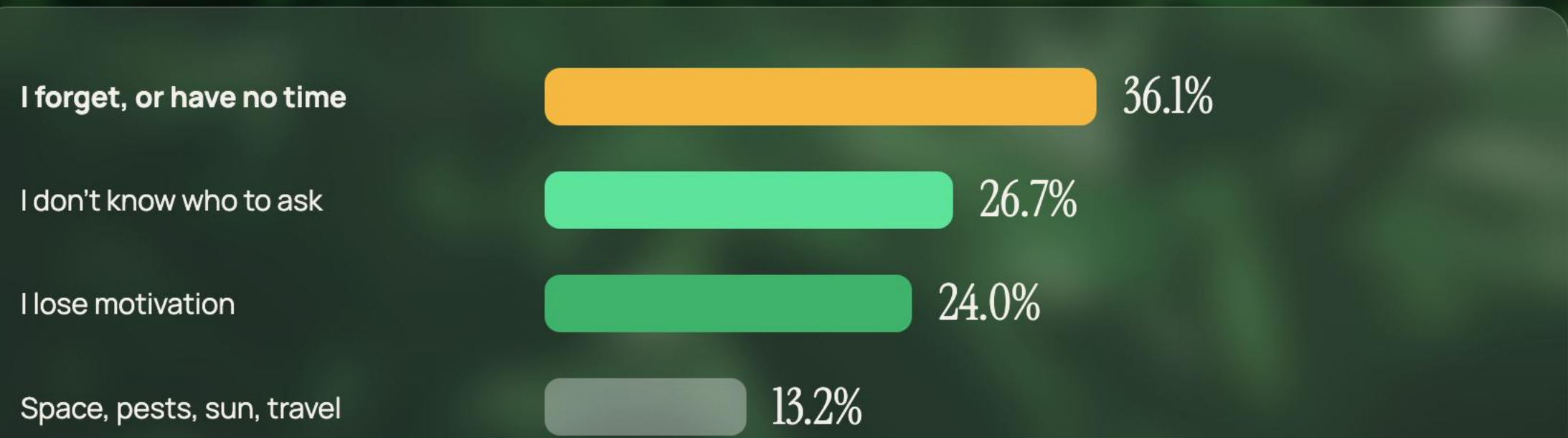
stopped growing or never started, though most want to



Interest is high: 70.9% rate it 4 or 5 out of 5.

WHY PLANTS DIE

The barrier is time and knowledge, not space



When unsure, 35.2% search Google and 16.3% do not know where to look at all.

OUR RESEARCH · 3,043 RESIDENTS

We measured the problem before building anything

What we asked	Result
Interest in growing at home (4–5 of 5)	70.9%
Growing plants right now	44.8%
Stopped, or never started	51.2%
Top barrier: forget or no time	36.1%
Would most value a diagnosis	34.0%
Likely to use the app (4–5 of 5)	79.7%

3,043

people answered our survey.

A sample this large means every result is accurate to within about ±1.8 percentage points.

THE HIDDEN COST

Every dead plant wastes more than a plant



Water

Poured into pots that fail, or onto the surface where it evaporates before the roots reach it.

Fertiliser

Given by guesswork, washed out of the pot and into the drains.

Soil, pots and seedlings

Bought, used once and thrown away when the plant dies.

A grower

The biggest loss: 23.6% of residents tried, failed and stopped growing.

OUR IDEA

A plant that can ask for what it needs, and get it, even when you forget.



THE SYSTEM

Four parts, one closed loop



Probe
Moisture, N·P·Mg and copper EC sensing in the root zone; feeds through the tip.

Leaf patch
Copper electrodes read stem signals linked to stress.

Module
ESP32-S3, DHT11 and two solar panels beside the pot.

Pump station
Two 5 V pumps for water and fertiliser, one tube.

HOW ROOTSTAKE WORKS

Sense, think, act, explain: one closed loop



1 · SENSE

- Probe: soil moisture at the roots
- Probe: N, P and Mg sensors, copper EC bands
- Leaf patch: stem electrical signals
- DHT11: air temperature and humidity



2 · THINK

- ESP32-S3 module calibrates every reading
- Works out how fast the soil is drying
- Forecasts the next watering
- Decides: water, feed, or wait



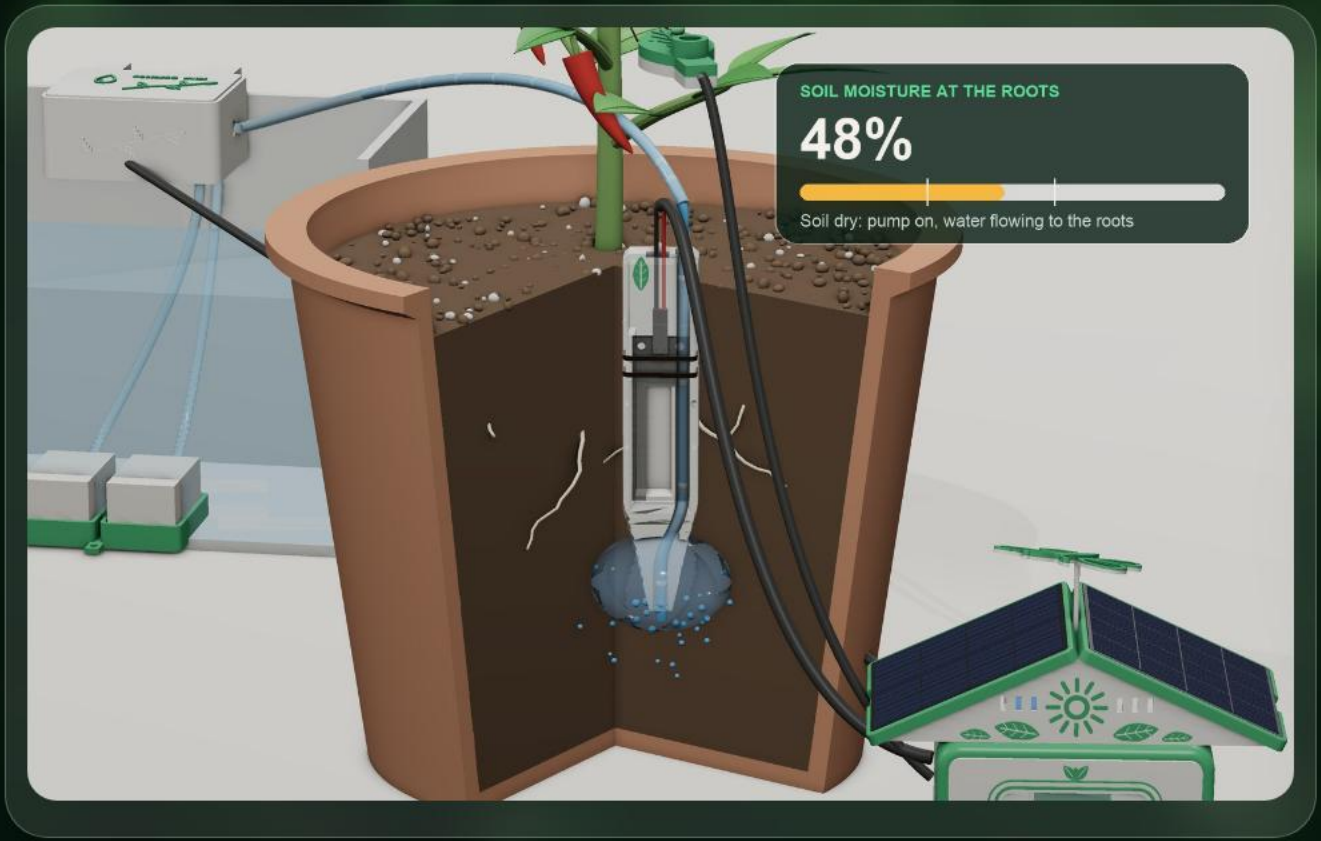
3 · ACT

- Relay switches pump 1 (water) or pump 2 (feed)
- Y-connector merges both into one tube
- Tube exits at the probe tip, in the root zone
- Stops at the plant's upper moisture limit

4 · EXPLAIN Bluetooth sends readings to the ROOTED SG app; Mojo turns them into one plain instruction. Solar powered.

SEE IT WORK

Watch the water reach the roots



1 · Soil reads dry
Moisture falls below the plant's lower limit.

2 · Pump on
Water travels down the hidden tube to the probe tip.

3 · Roots drink
A wet zone spreads in the root zone, below the surface.

4 · Pump off
At the upper limit the pump stops. No waste, no overwatering.

A DAY IN THE LIFE OF THE MODULE · EXAMPLE

What RootStake does while you are busy



Morning

Sun up, panels charging

The module wakes on solar power and starts its readings.



Every hour

Reads the plant

Moisture, nutrients, stem signals and air are logged and averaged.



Afternoon

Heat raises demand

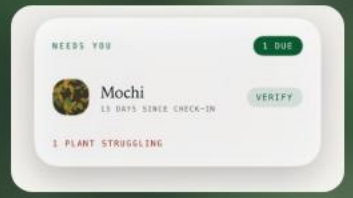
Hot, dry air speeds up drying, so the next watering is forecast earlier.



When dry

Waters at the roots

Below the lower limit, the pump runs until the soil reaches the upper limit.



Evening

One line in the app

Mojo sums up the day: watered once, nutrients fine, plant healthy.

THE PROBE

Sensing and feeding at the roots



Moisture fork
Measures water where roots actually drink.

N · P · Mg sensors
Track the nutrients plants most often lack.

Centre-tip outlet
Water and feed arrive below the surface, only when needed.

Built to go in
Chisel point with small saw teeth for potting soil.

THE MODULE

A solar module that looks after itself





ESP32-S3

Reads every sensor, runs the watering rule, talks to the app over Bluetooth.

Shaded DHT11

Air temperature and humidity under a vented roof.

2 × 70 mm solar

No socket, no charging.

ROOTED SG APP · IOS AND ANDROID

Every barrier from the survey has an answer



I forget (36.1%)
RootStake waters and feeds automatically.

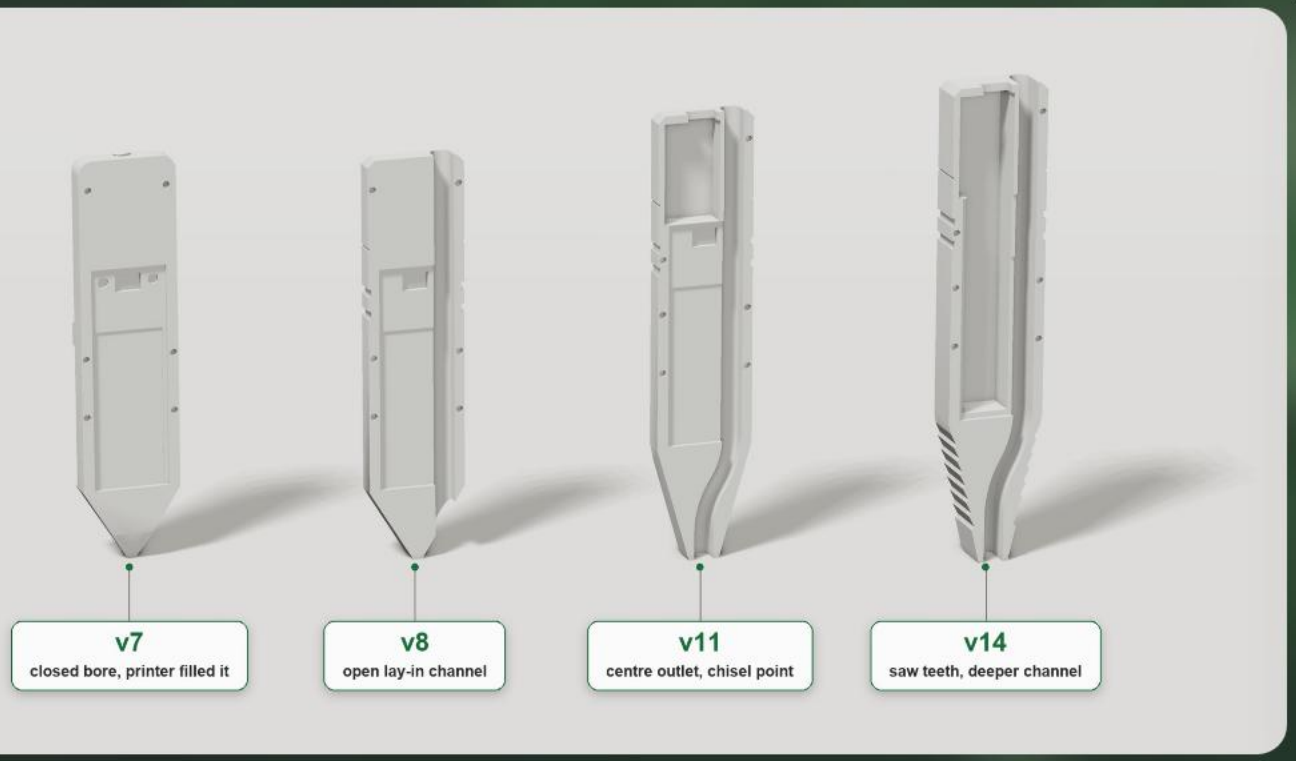
What is wrong? (34%)
Plant Doctor reads this plant's own data.

No one to ask (26.7%)
Mojo, the app's companion, explains in plain words.

I lose motivation (24%)
Missions, lessons and verified progress.

EVIDENCE OF REAL WORK

Fourteen probe designs, each failure a new rule



v7 printed with plastic inside its own sealed channel.

v8 switched to an open channel the tube lays into.

v11 moved the outlet to the centre and added a chisel point.

v14 added saw teeth and a deeper channel.

FIELD TEST

We ran RootStake for 15 days on 4 plants

4 / 4

plants healthy at the end

15

days of automatic watering and feeding

What comes next

The loop works. Next we prove how much better it is than hand watering.

Next: at least 30 plants, half hand-watered as controls, measuring survival, water and fertiliser per plant.

NUTRIENT SENSING · ACCURACY TARGETS

The accuracy we are aiming for, and how we will prove it

Nitrogen (N)

≥ 80%

agreement with lab low / OK / high classes

Phosphorus (P)

≥ 80%

agreement with lab low / OK / high classes

Magnesium (Mg)

≥ 80%

agreement with lab low / OK / high classes

Readings vs lab

r≥0.8

correlation between RootStake and lab values

TARGETS, NOT RESULTS Measured in the 30-plant trial: every RootStake reading compared with a laboratory soil test of the same sample.

LIMITATIONS AND NEXT EXPERIMENTS

What we still need to prove, and how



Small, short field test

15 days, 4 plants, no controls.

Next: 30+ plants with hand-watered controls.

Sensor drift

Resistive forks corrode over time and readings drift.

Next: log drift across the trial and recalibrate with dry and wet references.

Nutrient readings

N, P and Mg readings are not yet benchmarked.

Next: compare with laboratory soil tests during the trial.

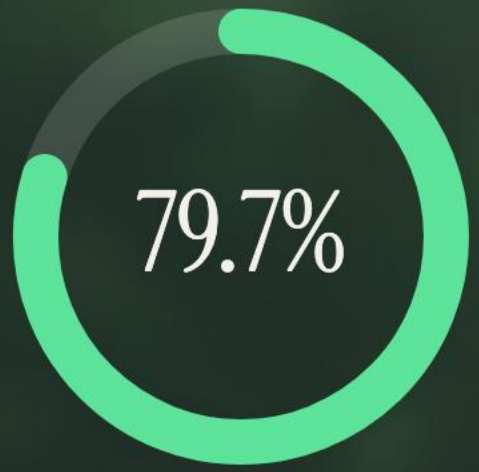
Survey sample

Self-selected online respondents.

Next: repeat with school and community-garden groups.

DEMAND

Residents want this

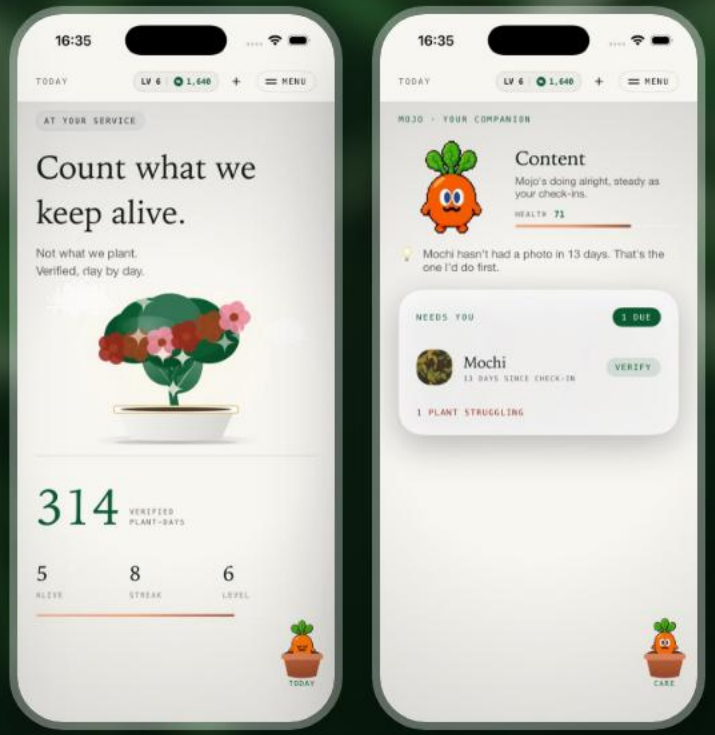


likely to use ROOTED SG (4 or 5 out of 5)

8

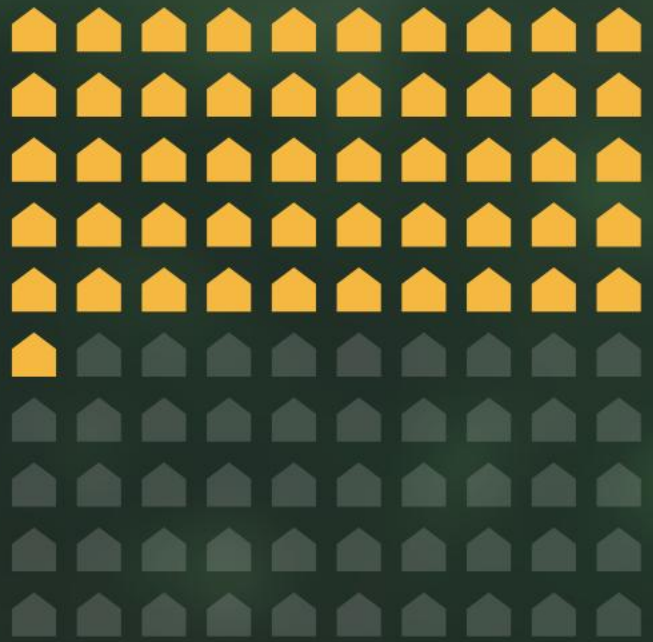
people out of 3,043 rated it 1 or 2.

40.1% rated it 4 and 39.6% rated it 5.



IF OUR SURVEY REFLECTS SINGAPORE

Half a million homes want to grow food and cannot



~500k

51 of every 100 homes want to grow. Applied to 1M+ HDB homes: an estimate, not a measurement.

Half a million micro-farms

Corridors, balconies and windowsills that already exist. No new land needed.

A national target

Food Story 2 wants local vegetables to rise from 8% to 20% by 2035.

Cheaper as it grows

One module serves many probes, so cost per plant keeps falling.

ENVIRONMENTAL IMPACT AND SCALE

Less water, less runoff, less waste



Less water wasted

Waters only when the soil is dry, below the surface where less evaporates.

Less fertiliser runoff

Small doses at the roots instead of washing out of pots.

Less waste, more growers

Surviving plants mean fewer seedlings and soil thrown away.

One module, many probes

Cost per plant falls as a garden grows: homes, schools, community plots.

THE VISION

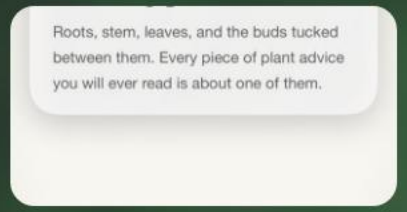
From one balcony to a city that grows its own food



TODAY

Homes

Working prototype; 15-day test on 4 plants, all healthy.



NEXT

School gardens

A pilot where students learn plant science from live data.



THEN

Community plots

Multi-probe modules looking after whole beds of crops.



FUTURE

A living city map

Anonymous plant-health data showing what grows best where, for growers and planners.

Every plant kept alive: water saved, fertiliser kept out of drains, food grown at home.

ROOTSTAKE BY TEAM ROOTEDSG

Grow with the space you have.

51.2% of residents want plants they do not have. We want every one of them to succeed.

Next: 30-plant controlled trial

School-garden pilot

Multi-probe module

